The one-loop twisted differential along a rank-one supercharge compared with the holomorphic twist at the level of classes

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Sources: [BK] arXiv:2512.07771v3, (3.1)–(3.70); [ESW] arXiv:2002.10517, Thm 10.9; [AL] the All-loop article (reader `127.0.0.1:8778`, Appendix B, blocks `b-01003`–`b-01027`; (3.1) `eq-A-28`, (3.2) `eq-1-1`, (3.5) `eq-B-8`, (3.6) `eq-B-10`, (7.1) `eq-2-3`, (7.4) `eq-Qnzero`, (7.6) `eq-2-26`, (7.7) `eq-2-27`, (B.1) `eq-A-20`, (B.6) `eq-A-13`, (B.7) `eq-A-15`, (B.9) `eq-A-16`). Manuscript: Lemma `c8-ev`, Def `c8-twisted`, Thm `c8-twist`, Rem `c11-twist`, Thm `c11-twistentries`, Thm `c11-twistcheck`, Rem `c12-twistcmp`. Code: `step2/next/twist/src.nosync`. "WS" marks the Weinberg–Srednicki two-component form (Definition `c8-twocomp`, $\sigma^m = (\vec{\sigma}, i)$, $\bar{\sigma}^m = (-\vec{\sigma}, i)$, $\varepsilon^{12} = \varepsilon^{\dot{1}\dot{2}} = 1$).

1 Setup

Definition 1 (Supercharge, twisted differential)**.** $\xi = u \otimes z \in S_+ \otimes W$, $u' \in S_+$ with $(u', u) = 1$, $z = e_4$; WS: ${\xi_\alpha}^i = u_\alpha z^i$, $\bar{\xi} = 0$, $u'^\alpha u_\alpha = 1$.

$$Q^\xi = r_\xi(\Lambda)|_{q=1} = Q_0^\xi + \hbar Q_1^\xi, \qquad Q_0^\xi = \{S^\xi, -\}, \qquad S^\xi = S + S^{(1)}(\xi) - \tfrac{1}{2}S^{(2)}(\xi,\xi),$$

$$\hbar Q_1^\xi = \hbar L_1^{(0)} + \hbar L_1^{(1)}(\xi) - \tfrac{1}{2}\hbar L_1^{(2)}(\xi,\xi) = \sum_{k=0}^{2} \Upsilon^{(k)} + (\text{one-body terms}),$$

$\Upsilon^{(k)}$ the ε^0 coefficients of the defect cycles with one special edge $K_\varepsilon - K_0$ and k vertices $S^{(1)}(\xi)$, each with coefficient 1 (Lemma `c8-ev`: $r_\xi W(P,J) = W(P, r_\xi J)$; the $-\frac{1}{2}$ of arity two is the $\frac{1}{2!}$ of two vertices at distinct chain positions times the sign $(-1)^{k(k-1)/2}$). $S^{(2)}(\xi,\xi)$ has two λ^* legs and sits on no cycle.

Definition 2 (Holomorphic side)**.** [AL, (B.6)], [AL, (B.7)], [AL, (7.4)] at $n = 1$:

$$C^A = ig\,\mathsf{c}^A + \theta_I \gamma^{IA} - \tfrac{1}{2}\varepsilon^{IJK}\theta_I\theta_J\beta_K^A + \tfrac{\theta_1\theta_2\theta_3}{ig}\mathsf{b}^A, \qquad Q_0 C^A = \tfrac{i}{2}{C^A}_{BC}C^B C^C,$$

$$Q_1\big|_{\text{jets}=0} = \frac{1}{128\pi^2}\int d^3\theta\; C_{ADR}C_{BRE}\,(\partial_{+\dot{a}}C^D)(\partial_+{}^{\dot{a}}C^E)\,\frac{\delta}{\delta C^A(\theta)}\frac{\delta}{\delta C^B(\theta)}.$$

[BK, (3.52)]: $C_{BK} = \mathsf{c} + \theta_I\gamma^I + \frac{1}{2}\varepsilon^{IJK}\theta_I\theta_J\beta_K + \theta_1\theta_2\theta_3\mathsf{b}$, $Q_0 C_{BK} = \frac{1}{2}[C_{BK}, C_{BK}]$, entries [BK, (3.64)–(3.69)] with the factor $\frac{1}{2}$ of [BK, fn 23].

Definition 3 (Normal form of a pair entry)**.** For letters X^A, Y^B and outputs U, V: $\langle U \cdot V\rangle := f_{ACD}f_{BCE}\,\partial_{\dot{\alpha}}U^D\partial^{\dot{\alpha}}V^E$ (U attached to X). In the engine, a chain order (Y_1, Y_2) carries $(1 \otimes \mathrm{ad}_{Y_2}\mathrm{ad}_{Y_1})(-,-)^{-1}$; with $\partial_k := d_k \cdot \partial$ (d_1, d_2 the null vectors of $u' \otimes S_-$),

$$\mathcal{S}[U,V] := F[\partial_1 U, \partial_2 V] - F[\partial_2 U, \partial_1 V],$$

$F[P,R]$ the functional of chain order (P next to X). $\mathcal{S}[U,V] \leftrightarrow \langle U\cdot V\rangle$ up to one factor common to all entries. WS: $\partial_{\dot{\alpha}} = u'^\alpha \sigma^m_{\alpha\dot{\alpha}}\partial_m$, $\langle U\cdot V\rangle = f_{ACD}f_{BCE}\,\varepsilon^{\dot{\alpha}\dot{\beta}}\partial_{\dot{\alpha}}U^D\partial_{\dot{\beta}}V^E$.

2 The transport map

Lemma 4 (Free twisted differential)**.** *On plane waves, $Q^{\xi}_{\text{free}} = Q + \xi. + \kappa\,\xi\xi^{T}|_{\lambda\leftarrow\lambda^*}$ is nilpotent iff $\kappa = \frac{1}{2}$, i.e. $\{-\frac{1}{2}S^{(2)}(\xi,\xi),-\}$ with $S^{(2)}(\xi,\xi) = -\frac{1}{2}\int((\xi,\lambda^*),(\xi,\lambda^*))$.* WS*: $-\frac{1}{2}S^{(2)} = \frac{1}{4}\int(u_\alpha z^i \lambda^{*\alpha}{}_i, u_\beta z^j \lambda^{*\beta}{}_j)$.*

Proof. $Q^2 = 0$, $Q\,\xi. + \xi.\,Q = 0$ (gate E0), $(\xi.)^2 + \kappa(QS_2 + S_2Q) = 0$ solved entrywise: unique $\kappa = \frac{1}{2}$ (`freeq.py`). □

Proposition 5 (Single-letter cohomology)**.** *Let $t = d + 2w$ (degree d, Weyl weight w) and fix the charges of $\mathfrak{so}(V) \oplus \mathfrak{so}(W^{\perp}_{\mathbb{R}})$ modulo the weight q_ξ of ξ. In each sector below $H(Q^{\xi}_{\text{free}})$ is one-dimensional, with the representatives*

letter	*index-free (engine realisation)*	WS	(w,d)
c	c	c	$(0,1)$
γ^I	$\iota_{w_I}A,\ \rho(w_I)\xi = 0$	A_{I4}	$(1,0)$
β_I	$(u\otimes\tau_I)$*-component of* $\lambda|_{S_+\otimes W}$	$u'^{\alpha}\lambda_\alpha{}^I$	$(\frac{3}{2},0)$
$\partial_k\mathsf{c}$	$\lambda|_{S_-\otimes W^*}$ *along* $(s_k;-z)$ $\sim$ $\partial_k c$	$\bar\lambda_{\dot\alpha 4} \sim u'^{\alpha}\sigma^m_{\alpha\dot\alpha}\partial_m c$	$(\frac{3}{2},0)\sim(1,1)$
b	$\frac{1}{2}F(d_1,d_2) + (u'\otimes z,\lambda^*)$	$u'^{\alpha}F_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta u'_\beta + \kappa_{\mathsf{b}}\,u'_\alpha\lambda^{*\alpha}{}_4$	$(2,0)+(\frac{5}{2},-1)$
$\partial_k X$	∂_k(*rep of* X)	$u'^{\alpha}\sigma^m_{\alpha\dot\alpha}\partial_m(\cdots)$	

with the relations (exact, engine realisation; λ-components on dual weight rows)

$$Q^{\xi}_{\text{free}}(\iota_{d_1}A) = -\partial_1 c + 32i\,\bar\lambda_{(1)}, \qquad Q^{\xi}_{\text{free}}(\iota_{d_2}A) = -\partial_2 c - 32i\,\bar\lambda_{(0)},$$

so $\partial_1 c \equiv 32i\,\bar\lambda_{(1)}$, $\partial_2 c \equiv -32i\,\bar\lambda_{(0)}$; antiholomorphic derivatives of c are exact. WS*: $Q^{\xi}(u'^{\alpha}\sigma^m_{\alpha\dot\alpha}A_m) = \partial_{\dot\alpha}c + 2i\bar\lambda_{\dot\alpha 4}$, i.e. $\partial_{\dot\alpha}c \equiv -2i\,\bar\lambda_{\dot\alpha i}z^i$. The element (ξ,λ^*) (*WS*: $u_\alpha\lambda^{*\alpha}{}_4$) completed by $\frac{i}{2}(\partial\cdot A)$-terms is Q^{ξ}_{free}-exact; it does not represent b.*

Proof. Exact linear algebra on the charge sectors (`sector.py`, `t3.py`, `t4.py`, `t8.py`): the b sector $(\chi_{\mathsf{b}}, t=4)$, $\chi_{\mathsf{b}} = d_1 + d_2$, has basis $\{y_{-d_1}\iota_{d_2}A,\ y_{-d_2}\iota_{d_1}A,\ (u'\otimes z,\lambda^*)\}$, $t-1$ part empty, and kernel spanned by $i\,y_{-d_1}\iota_{d_2}A - i\,y_{-d_2}\iota_{d_1}A + (u'\otimes z,\lambda^*) = \frac{1}{2}F(d_1,d_2) + (u'\otimes z,\lambda^*)$ $(\partial_{n_w} \leftrightarrow 2i\,y_{-w})$. The charge-0, $t=4$ sector has $\dim Z = \dim B = 1$, $Z \ni (\xi,\lambda^*) + \frac{i}{2}(\cdots)$. □

Remark 6 (AL, (3.3))**.** (`eq-field-form-dictionary`): $\mathsf{b}_A = \frac{1}{4}\delta_{AB}f^B_{++} - \frac{i}{2}\lambda^{\ddagger -}_A + \cdots$, the same structure (field strength of the $u'u'$ component plus the gaugino antifield along u'); [AL, fn 16]: $[\bar\lambda_{\dot a}] = \frac{1}{2}[\partial_{+\dot a}\mathsf{c}]$, the analogue of $\partial_{\dot\alpha}c \equiv -2i\bar\lambda_{\dot\alpha 4}$.

Proposition 7 (Tree level)**.** *In the vector-field convention of Q^{ξ}_{free}, $Q^{\xi}_0\lambda = [c,\lambda] + \frac{1}{2}\rho(F_A)\xi$ and $Q^{\xi}_0\lambda^* \ni \not{A}_A\lambda$. On the representatives,*

$$Q^{\xi}_0\beta_{(i)} = [c,\beta_{(i)}] + c_{i;jk}[\gamma_{(j)},\gamma_{(k)}], \qquad (c_{0;12},c_{1;02},c_{2;01}) = (-\tfrac{1}{2},-\tfrac{i}{2},\tfrac{i}{2}),$$

$$Q^{\xi}_0\mathsf{b} \ni [c,\mathsf{b}] + \textstyle\sum_i d_i[\gamma_{(i)},\beta_{(i)}], \qquad (d_0,d_1,d_2) = (-16i,16,16), \qquad \frac{d_i\,c_{i;jk}}{\varepsilon_{ijk}} = 8i \quad (i=0,1,2).$$

Hence Φ intertwines Q^{ξ}_0 with [BK, (3.49)–(3.50)],

$$\mathsf{c}_{BK} = c, \qquad \gamma^I_{BK} = \gamma_{(I-1)}, \qquad \beta_{(i)} = a_i\,\beta_{BK,i+1}, \quad (a_0,a_1,a_2) = (-\tfrac{1}{2},\tfrac{i}{2},\tfrac{i}{2}),$$
$$\mathsf{b} = -8i\,\mathsf{b}_{BK}, \qquad \partial_k \mapsto \partial_{\dot\alpha=k},$$

up to the one-parameter rescaling $s_t : \gamma \mapsto t\gamma$, $\beta \mapsto t^2\beta$, $\mathsf{b} \mapsto t^3\mathsf{b}$, which preserves Q_0 and rescales Q_1 by t^{-3} (the normalisation of $\hbar_{\text{tw}}$). The uniform value of $d_i c_{i;jk}/\varepsilon_{ijk}$ is the statement $Q_0\mathsf{b} = [\mathsf{c},\mathsf{b}] - \sum_I[\gamma^I,\beta_I]$ with one coefficient. WS*: $Q^{\xi}_0(u'^{\alpha}\lambda_\alpha{}^I) = [c,u'^{\alpha}\lambda_\alpha{}^I] + \frac{1}{4}[A^{Ik},A_{k4}]$, $Q^{\xi}_0\bar\lambda_{\dot\alpha i} = [c,\bar\lambda_{\dot\alpha i}] + \frac{i}{2}u^\alpha\sigma^m_{\alpha\dot\alpha}D_m(A_{ij}z^j)$.*

Proof. $F_{MN} = [A_M, A_N]$ on $W_{\mathbb{R}}^{\perp}$ and $d_A = d + [A, -]$ give the quadratic parts from the free arrows; the gauge terms follow from $(Q_0^{\xi})^2 = 0$ ($Q_0^{\xi} c = \frac{1}{2}[c, c]$). Projection on $\beta_{(i)}$ and on $(u' \otimes z, \lambda^*)$ (`q0tree.py`). Labels: $Q_0^{\xi}\beta_{(i)} \ni [\gamma_{(j)}, \gamma_{(k)}]$ only for $\{j,k\} = \{0,1,2\} \setminus \{i\}$, and $\beta_{(i)}\gamma_{(j)}$ has the charge of $\partial_1 \mathsf{c}\, \partial_2 \mathsf{c}$ (mod q_ξ) only for $i = j$ (`t5.py`). □

Proposition 8 (User normalisation versus BK)**.** *With $C_{BK} = i\,C$ (so $\mathsf{c}_{BK} = -g\mathsf{c}$, $\gamma_{BK} = i\gamma$, $\beta_{BK} = -i\beta$, $\mathsf{b}_{BK} = \mathsf{b}/g$), $[AL, (7.4)] = \frac{1}{32\pi^2}\,\Phi_{AL}^*\big([BK, (3.64)\text{–}(3.69)]\big)$ entry by entry; in particular $[AL, (7.6)]$, $[AL, (7.7)]$ are reproduced by the Grassmann evaluation of $[AL, (7.4)]$:*

pair	$[AL, (7.4)]:\ C_{A'PR}C_{B'RQ}W_1^P \cdot W_2^Q,\ \times 64\pi^2$	$[BK]\ \langle\cdot\rangle$
$\gamma^1\beta_1$	$\mathsf{c}.\mathsf{c}:\ -g^2$	$\frac{1}{2}\ \mathsf{cc}$
$\beta_1\beta_2$	$\mathsf{c}.\gamma^3:\ -ig, \quad \gamma^3.\mathsf{c}:\ ig$	$\frac{1}{2}\varepsilon(\mathsf{c}\gamma - \gamma\mathsf{c})$
bc	$\mathsf{c}.\mathsf{c}:\ g^2$	$\frac{1}{2}$
$\mathsf{b}\gamma^1$	$\mathsf{c}.\gamma^1:\ g^2, \quad \gamma^1.\mathsf{c}:\ -g^2$	$\frac{1}{2}(\mathsf{c}\gamma - \gamma\mathsf{c})$
$\beta_1\mathsf{b}$	$\mathsf{c}.\beta_1,\ \beta_1.\mathsf{c}:\ g^2; \quad \gamma^2.\gamma^3:\ ig,\ \gamma^3.\gamma^2:\ -ig$	$\frac{1}{2}(\beta\mathsf{c} + \mathsf{c}\beta + \varepsilon\gamma\gamma)$
bb	$\mathsf{c}.\mathsf{b}:\ g^2,\ \mathsf{b}.\mathsf{c}:\ -g^2,\ \gamma^I.\beta_I:\ g^2,\ \beta_I.\gamma^I:\ -g^2$	$\frac{1}{2}(\mathsf{cb} - \mathsf{bc} + \beta\gamma - \gamma\beta)$
$\gamma\gamma,\ \mathsf{cc},\ \beta\mathsf{c},\ \gamma\mathsf{c},\ \beta_1\gamma^2$	0	0

Proof. `holo.py`: Grassmann algebra on $\theta_{1,2,3}$ and the component letters, $\delta/\delta C^A$ of [AL, (B.9)], $\int d^3\theta\, \theta_1\theta_2\theta_3 = 1$. Under $C_{BK} = iC$, $Q_0 C = \frac{i}{2} C^A{}_{BC} C^B C^C$ becomes $\frac{1}{2}[C_{BK}, C_{BK}]$; the transported BK coefficient of each entry is $b_E\, \rho_U \rho_V / (\rho_X \rho_Y)$ with $(\rho_{\mathsf{c}}, \rho_\gamma, \rho_\beta, \rho_{\mathsf{b}}) = (-g, i, -i, 1/g)$ and $f_{ACD} f_{BCE} = -f_{APR} f_{BRQ}$; the ratio to [AL] is $1/(32\pi^2)$ in all fourteen channels. □

3 What is invariant

Theorem 9 (Class)**.** *Let $\mathcal{F}$ be the space of local operators, graded by the charges of the commutant of ξ (finite-dimensional sectors). Then:*

(i) $[Q_0^{\xi}, Q_1^{\xi}] = 0$*, and the class* $[Q_1^{\xi}] \in H\big(\operatorname{End}\mathcal{F}, \operatorname{ad}_{Q_0^{\xi}}\big) \cong \operatorname{Hom}\big(H(Q_0^{\xi}), H(Q_0^{\xi})\big)$ *is invariant under: the field redefinitions and conjugations of Prop* `c8-laws` *(a); the finite parts ψ of the operators of every order (Def* `c8-renorm`*), including their arity components; the action-side free directions ($\theta_V^{\xi}, \theta_{\perp}^{\xi}$ of $\hbar F_1^{(1)}(\xi)$, $\hbar F_1^{(0)}$, $\hbar F_1^{(2)}(\xi,\xi)$); the coupling direction; the outer direction G_B; the homotopy data (p, h) of the transport.*

(ii) $\mathbb{C}\,Q_0^{\xi} \subset \operatorname{Im}\operatorname{ad}_{Q_0^{\xi}}$*: $Q_0^{\xi} = -[Q_0^{\xi}, N_t]$ with N_t the derivation counting t. Hence "modulo $\mathbb{C}\,S^{\xi}$" is contained in "modulo* $\operatorname{Im} Q_0^{\xi}$*".*

(iii) Two quantisations of the twisted theory differing by a Q_0^{ξ}-closed degree-0 local functional D have $[Q_1^{\xi}]$ differing by $[\{D, -\}]$; it vanishes for $D \in \mathbb{C}S^{\xi} + \operatorname{Im} Q_0^{\xi}$.

Proof. (i) Theorem `c8-twist`: each change is $Q_1^{\xi} \mapsto Q_1^{\xi} + [Q_0^{\xi}, \phi]$ or $Q_1^{\xi} - aQ_0^{\xi}$; for ψ of arbitrary order, r_ξ of the conjugation by $1 + \hbar\psi$ gives $[Q_0^{\xi}, r_\xi\psi]$; the free directions enter through $-\{\hbar F_1^{(k)}, -\}$, brackets with Q_0^{ξ}-closed functionals, which are $[Q_0^{\xi}, \{\chi, -\}]$ for exact ones and are the listed directions otherwise; two choices of homotopy data give transferred differentials related by an L_∞ quasi-isomorphism, $\tilde{Q}_1 \mapsto \tilde{Q}_1 + [\tilde{Q}_0, \phi_h]$. Over $\mathbb{C}$, $H(\operatorname{End} K) \cong \operatorname{End} H(K)$ for each finite complex K. (ii) $[N_t, Q_0^{\xi}] = Q_0^{\xi}$. (iii) $\{S^{\xi}, -\} = Q_0^{\xi}$, $\{Q_0^{\xi}\chi, -\} = \pm[Q_0^{\xi}, \{\chi, -\}]$. □

Theorem 10 (Entries in one-loop-local schemes)**.** *Restrict the changes of Theorem 9 to those whose operator parts act on at most two letters (finite parts ψ of order ≤ 2, $so(V) \oplus so(W_{\mathbb{R}}^{\perp})$-covariant), as all one-loop graphs do. For each pair (X, Y) of holomorphic letters and outputs $\partial U, \partial V$, $U, V \in \{\mathsf{c}, \gamma, \beta, \mathsf{b}\}$, the coefficient of $\langle U \cdot V\rangle$ in the transported $Q_1^{\xi}(XY)$ is invariant. Under a finite part of order ≥ 3 it is not (Rem* `c11-twist`*: $\psi(W) = y \operatorname{tr} \partial\mathsf{c} \cdot \partial\mathsf{c}$ moves the Konishi coefficient by $-3y$).*

Proof. On holomorphic letters $\tilde{Q}_0$ is $1 \to 2$. For a two-body $\tilde{\psi}$, $[\tilde{Q}_0, \tilde{\psi}]_{2\to 2} = \tilde{Q}_0\tilde{\psi}_{2\to 1} - \tilde{\psi}_{2\to 1}\tilde{Q}_0$ ($\tilde{\psi}_{3\to 2} = 0$; the perturbation-lemma terms $p\,\psi h Q_{\rm int} i$, $p\,Q_{\rm int} h \psi\, i$ have ≥ 3 outputs on two inputs). $\tilde{\psi}_{2\to 1}(XY)$ must carry the charges of XY: t, $H_1 = 3\#\partial + 2s$ ($s(\mathsf{c},\gamma,\beta,\mathsf{b}) = (0,1,2,3)$), SU(3) and the dotted J_R; the only letters with the first three are $\partial_{(\dot\alpha}\partial_{\dot\beta)}X'$, dotted triplets, while XY is a dotted singlet. $\tilde{\psi}_{2\to 1}\tilde{Q}_0(XY)$ keeps an undifferentiated spectator in $\{\mathsf{c},\gamma,\beta,\mathsf{b}\}$, absent from $\partial U\,\partial V$; one-body changes keep a spectator. Charges (t, H_1): $\mathsf{c}(1,0)$, $\gamma(2,2)$, $\beta(3,4)$, $\mathsf{b}(4,6)$, $\partial(2,3)$. □

4 The comparison

Theorem 11 (Pair entries)**.** *In the scheme $\psi = 0$ with the computed $\hbar F_1$, the transported entries of Q_1^ξ (all arities, $\mathsf{b} = \frac{1}{2}F(d_1,d_2) + (u'\otimes z, \lambda^*)$, outputs projected by p) are, in units $\hbar/(16\pi^2)$ and the structures $\mathcal{S}$ (engine realisation, flavour labels $(0,1,2) \leftrightarrow (1,2,3)$):*

entry	*channel*	*ours o_E*	*[BK] b_E*	$r_E = o_E \frac{a_U a_V}{a_X a_Y} b_E^{-1}$
$\beta_{(0)}\gamma_{(0)}$ [BK (3.66)]	$\mathcal{S}[\mathsf{c},\mathsf{c}]$	$\frac{i}{16}$	$\frac{1}{2}$	$-\frac{i}{4}$
$\beta_{(1)}\gamma_{(1)},\ \beta_{(2)}\gamma_{(2)}$	$\mathcal{S}[\mathsf{c},\mathsf{c}]$	$\frac{1}{16},\ \frac{1}{16}$	$\frac{1}{2}$	$-\frac{i}{4}$
$\beta_{(0)}\gamma_{(1)}$		0	0	
$\beta_{(0)}\beta_{(1)}$ [BK (3.67)]	$\mathcal{S}[\mathsf{c},\gamma_{(2)}],\ \mathcal{S}[\gamma_{(2)},\mathsf{c}]$	$-\frac{1}{32},\ \frac{1}{32}$	$\frac{1}{2},\ -\frac{1}{2}$	$-\frac{i}{4}$
$\beta_{(1)}\beta_{(2)}$	$\mathcal{S}[\mathsf{c},\gamma_{(0)}]$	$\frac{i}{32}$	$\frac{1}{2}$	$-\frac{i}{4}$
bc [BK (3.64)]	$\mathcal{S}[\mathsf{c},\mathsf{c}]$	-1	$\frac{1}{2}$	$-\frac{i}{4}$
$\mathsf{b}\gamma_{(0)},\ \mathsf{b}\gamma_{(1)}$ [BK (3.68)]	$\mathcal{S}[\mathsf{c},\gamma],\ \mathcal{S}[\gamma,\mathsf{c}]$	$-1,\ 1$	$\frac{1}{2},\ -\frac{1}{2}$	$-\frac{i}{4}$
$\beta_{(0)}\mathsf{b}$ [BK (3.69)]	$\mathcal{S}[\beta_{(0)},\mathsf{c}],\ \mathcal{S}[\mathsf{c},\beta_{(0)}]$	$-1,\ -1$	$\frac{1}{2},\ \frac{1}{2}$	$-\frac{i}{4}$
	$\mathcal{S}[\gamma_{(1)},\gamma_{(2)}],\ \mathcal{S}[\gamma_{(2)},\gamma_{(1)}]$	$\frac{1}{2},\ -\frac{1}{2}$	$\frac{1}{2},\ -\frac{1}{2}$	$-\frac{i}{4}$
bb [BK (3.65)]	$\mathcal{S}[\mathsf{c},\mathsf{b}],\ \mathcal{S}[\mathsf{b},\mathsf{c}]$	$-1,\ 1$	$\frac{1}{2},\ -\frac{1}{2}$	$-\frac{i}{4}$
	$\mathcal{S}[\beta_{(0)},\gamma_{(0)}],\ \mathcal{S}[\gamma_{(0)},\beta_{(0)}]$	$-16i,\ 16i$	$\frac{1}{2},\ -\frac{1}{2}$	$-\frac{i}{4}$
$\gamma_{(0)}\gamma_{(1)},\ \beta_{(0)}\mathsf{c}$		0	0	

with $a_{\mathsf{c}} = a_\gamma = 1$, $a_{\beta_{(i)}} = a_i$, $a_{\mathsf{b}} = -8i$ (Proposition 7). Hence

$$\Phi \circ Q_1^\xi \circ \Phi^{-1} = -\tfrac{i}{4}\,\frac{\hbar}{16\pi^2}\,\mathcal{N}\,Q_1^{BK}$$

on every pair of holomorphic letters without derivatives, $\mathcal{N}$ the common conversion $\mathcal{S} \leftrightarrow \langle\cdot\rangle$, $\partial_k \leftrightarrow \partial_{\dot\alpha}$; under s_t the factor becomes $-\frac{i}{4t^3}$. Every $\log\frac{3}{2}$ cancels; every entry has the colour-order and dotted structure of [BK].

Proof. For each output class the sector basis and p (Proposition 5); for each pair of output basis elements with $p \neq 0$, the arity $k = 2(w_{\rm out} - w_{\rm in})$ is fixed per term of the input representatives, and the kernel is evaluated exactly (engine A defect cycles, $k = 0$: no vertex $S^{(1)}(\xi)$; $k = 1$; $k = 2$: two vertices, `qa2.py`), both chain orders, polynomial in the output null coordinates by exact interpolation (`twdriver.py`). The class bases are converted by $\partial_1\mathsf{c} \equiv 32i[\bar\lambda_{(1)}]$, $\partial_2\mathsf{c} \equiv -32i[\bar\lambda_{(0)}]$, $\partial_k X = 2i\,y_{-d_k}X$. Contributions in the class bases of `twdriver.projection` (d_1d_2 component;

pairs: chain order (U,V); (V,U); $L=\log\frac{3}{2}$, $\ell=\log 2$; F, λ^* the two terms of b):

entry	arity, part	(U,V); (V,U)
$\beta_{(0)}\gamma_{(0)}\to \mathsf{c}\,\mathsf{c}$	0	$64i$; $-64i$
	1	0; 0
$\beta_{(0)}\beta_{(1)}\to \mathsf{c}\,\gamma_{(2)}$	0	$\frac{4}{3}$; $\frac{4}{3}$
	1	$\frac{2}{3}$; $\frac{2}{3}$
$\mathsf{bc}\to \mathsf{c}\,\mathsf{c}$	0, F	$-512L-\frac{1280}{3}$; $512L+\frac{1280}{3}$
	0, λ^*	$512L-\frac{256}{3}$; $\frac{256}{3}-512L$
	1, F	$1024(L-\ell)-512$; $-1024(L-\ell)+512$
	1, λ^*	$-1024(L-\ell)$; $1024(L-\ell)$
	2, F	0; 0
$\mathsf{b}\gamma_{(0)}\to \mathsf{c}\,\gamma_{(0)}$	0, F	$64L+\frac{128}{3}$; $64L+32$
	0, λ^*	$\frac{16}{3}-96L$; $\frac{64}{3}$
	1, F	$32L+16$; $\frac{32}{3}-64L$
$\beta_{(0)}\mathsf{b}\to \beta_{(0)}\,\mathsf{c}$	0, F	$-64L-\frac{128}{3}$; $\frac{128}{3}$
	0, λ^*	$64L-\frac{32}{3}$; $\frac{64}{3}$
	1, F	$-\frac{32}{3}$; 0
$\beta_{(0)}\mathsf{b}\to \gamma_{(1)}\gamma_{(2)}$	0, λ^*	$-\frac{2}{3}$; $-\frac{2}{3}$
	1, F	$-\frac{4}{3}$; $-\frac{4}{3}$
$\mathsf{bb}\to \mathsf{c}\,\mathsf{b}$	0, $F\lambda^*\mid\lambda^*F$	$\frac{32}{3}$; 0 $\mid$ 0; $\frac{32}{3}$
	1, $FF\mid F\lambda^*\mid\lambda^*F$	$64L+32$; $64L+32$ $\mid$ 8; $\frac{32}{3}-64L$ $\mid$ $\frac{32}{3}-64L$; 8
	2, FF	$\frac{8}{3}$; $\frac{8}{3}$
$\mathsf{bb}\to \beta_{(0)}\gamma_{(0)}$	0, $F\lambda^*\mid\lambda^*F$	$\frac{64i}{3}$; $\frac{64i}{3}$ $\mid$ $\frac{64i}{3}$; $\frac{64i}{3}$
	1, FF	$\frac{64i}{3}$; $\frac{64i}{3}$

The arity-one rows with $\mathsf{b}\to F$ (doubled, $\mathsf{b}\leftrightarrow F(d_1,d_2)$) are the entries of Table `c11-twistentries`: $\beta\beta$: $\frac{2}{3}$; $\mathsf{b}\gamma$: $32+64L$, $\frac{64}{3}-128L$; $\beta\mathsf{b}\to\beta\,\partial\mathsf{c}$: $-\frac{64}{3}$, 0; $\beta\mathsf{b}\to\gamma\gamma$: $\mp\frac{8}{3}$; $\beta\gamma$: 0. The sums are the table of the theorem. □

Corollary 12 (Classes)**.** *In one-loop-local schemes the comparison is entry by entry (Theorem 10) and agrees with [BK, (3.64)–(3.69)] and with [AL, (7.4)] in all eight channels, with one constant. The Konishi entry is nonzero:*

$$Q_1^{\xi}(\beta_I\gamma^J)=\delta_I^J\,\big(-\tfrac{i}{4}\mathcal{N}\big)\tfrac{\hbar}{16\pi^2}\cdot\tfrac{1}{2}f_{ACD}f_{BCE}\,\partial_{\dot\alpha}\mathsf{c}^D\partial^{\dot\alpha}\mathsf{c}^E,$$

WS*:* $Q_1^{\xi}\big((u'^{\alpha}\lambda_{\alpha}{}^I)(A_{J4})\big)\ni\delta_J^I\,\bar\lambda_{\dot\alpha 4}\bar\lambda^{\dot\alpha}{}_4$ *through the arity-zero output* $\bar\lambda_{\dot\alpha 4}\,u'^{\alpha}\sigma_{\alpha}^{m\dot\alpha}\partial_m c$ *and* $\partial_{\dot\alpha}c\equiv-2i\bar\lambda_{\dot\alpha 4}$*. No entry is a genuine mismatch. The eight channels satisfy the relations [BK, (3.59)–(3.63)] with the same constant, so* Q_1^{ξ} *and* $\Phi^{-1}Q_1^{BK}\Phi$ *agree, up to that constant, on all pairs of underived letters; the entries with input derivatives [BK, (B.5)] are not computed here.*

Remark 13 (Manuscript Table `c11-twistentries`)**.** The table lists the arity-one kernels on one representative of each output. The arity-zero kernels with outputs $\partial_k c$ (ghost derivative) and, for b, the term $(u'\otimes z,\lambda^*)$ of the representative are absent there. With them: the Konishi entry is $\frac{i}{16}\mathcal{S}[\mathsf{c},\mathsf{c}]$, not 0; the $\log\frac{3}{2}$ terms and the colour-order asymmetries of $\mathsf{b}\gamma$ and $\beta\mathsf{b}\to\beta\,\partial\mathsf{c}$ cancel; the row "bc, $\mathsf{b}\leftrightarrow(\xi,\lambda^*)$" is the entry of an exact element. Thm `c11-twistentries` (i)–(iii) are statements about these representatives; Proposition `c11-plane` ($\gamma\gamma$) stands.

Remark 14 (Checks)**.** (a) Bilinearity and flavour covariance: $\beta_{(i)}\gamma_{(i)}$, $i=0,1,2$, and $\beta_{(0)}\beta_{(1)}$, $\beta_{(1)}\beta_{(2)}$, $\mathsf{b}\gamma_{(0)}$, $\mathsf{b}\gamma_{(1)}$ give the same r_E (the flavour normalisation enters only through $c_{i;jk}$). (b) Normalisation-free test: $r_{\beta\beta}r_{\mathsf{b}\gamma}=r_{\beta\gamma}r_{\beta\mathsf{b}\to\gamma\gamma}$ (letters balanced). (c) The arity-two kernels: 0 on $(F,\mathsf{c})\to(\bar\lambda,\bar\lambda)$, $\frac{8}{3}$ of 64 in $\mathsf{bb}\to\partial\mathsf{c}\,\partial\mathsf{b}$, required for $r=-\frac{i}{4}$. (d) Independent interpolation route (`assemble3`) reproduces bc. (e) Homotopy data: with the complement of Z in the $\partial\mathsf{b}$ sector reversed ($\partial\mathsf{b}$ read through ∂F instead of $\partial\lambda^*$, other kernels and arities), $\mathsf{bb}\to\partial\mathsf{c}\,\partial\mathsf{b}$ is again 64; 64 in both

dotted components. (g) The arity-one rows reproduce Table `c11-twistentries`. (f) Holomorphic side: [AL, (7.6)], [AL, (7.7)] reproduced by the Grassmann evaluation of [AL, (7.4)].

Remark 15 (Open)**.** The absolute value of the common constant (the factor t^3 of s_t, i.e. $\hbar_{\mathrm{tw}}$ against $\hbar\,\mathrm{tr}(\mathrm{adad})/(16\pi^2(-,-))$) requires the transported BV pairing on the Dolbeault components ([ESW, Thm 10.9] on full fields, [AL, (B.1)]); not computed. Entries with input derivatives ([BK, (B.5)]), three-letter classes and the lifting statements, and a second engine for the arity-zero and arity-two kernels, remain.